On the exact quantization of BCOV Theory on Calabi-Yau Manifolds

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October 15, 2026

Abstract

We announce the solution of constructing perturbative quantum BCOV invariants on arbitrary Calabi-Yau manifolds. In this note, we outline the construction and present an explicit solution. The quantization includes all descendant fields and all loop orders, and expresses the quantum BCOV interaction by an explicit integration over the moduli space of marked Riemann surfaces. This canonical solution provides a mathematical construction of all higher genus BCOV invariants on compact Calabi–Yau manifolds, which are conjecturally equivalent to all higher genus Gromov-Witten invariants on the mirror.

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Introduction

The Kodaira-Spencer theory of gravity was introduced by Bershadsky–Cecotti–Ooguri–Vafa [2] as the closed string field theory corresponding to the B-twisted topological string. The original formulation of BCOV theory was on Calabi-Yau manifolds of dimension 3. In [4], Costello and the first author extended Kodaira-Spencer gravity to Calabi-Yau manifolds of arbitrary dimension by incorporating with descendant fields, and formulated the general quantization framework. This is a quantization in Costello's effecitve BV formalism [3], and a solution of effective quantum master equation (which we call BCOV quantization) leads to a rigorous construction of higher genus B-model invariants on a Calabi-Yau manifold (see also [12]). In [4], a canonical BCOV quantization was shown to exist on elliptic curves by deformation theory. This was later explicitly constructed in [13], in which the full quantum BCOV invariants on elliptic curves were proven to be equivalent to all genus Gromov-Witten invariants on the mirror. In [5], a BCOV quantization was shown to exist locally on the flat space $\mathbb{C}^N$ by coupling with holomorphic Chern-Simons theory and anomaly cancellation argument. At one loop, this anomaly cancellation is the topological string analogue of the Green–Schwarz mechanism [6].

In this note, we present an explicit quantization of BCOV theory on any compact Calabi–Yau manifold in the Costello-Li framework [4]. The construction includes all descendants and is valid at every loop order. More precisely, let X be a closed connected Calabi–Yau manifold of complex dimension N, with holomorphic volume form Ω and a Kähler metric g satisfying (2.1). The field complex is $\mathrm{PV}(X)[[\zeta]][2]$, with differential $Q = \bar\partial - \zeta\partial_\Omega$. Write a field

$$A(\zeta) = \sum_{k\geq 0} \zeta^k A_k, \qquad A_k \in \mathrm{PV}(X)$$

A_0 is the primary field and A_k has descendant index k. We construct a local interaction (2.7)

$$\mathcal{J}_X = \sum_{g\geq 0} \hbar^g I_g.$$

I_0 is the genus-zero classical descendant BCOV interaction of Costello–Li [4]. The higher loop I_g's are local functionals consisting of holomorphic derivatives with coupling constants given by explicit integrations over the moduli space of marked Riemann surfaces (see below).

At every scale $L > 0$, let $I_X[L]$ be the effective functional that is built up from Feynman graph integrals with vertex $\mathcal{J}_X$ and propagator $P_0^L = \int_0^L \partial\bar\partial^* e^{-t[\bar\partial,\bar\partial^*]}dt$. Since BCOV theory is a holomorphic field theory, the above Feynman graph integrals are free of ultraviolet divergence and they are well-defined by Cauchy principal values [17, 18], without the need for singular ultraviolet counterterms. Theorem 2.1 proves that the resulting interaction satisfies the renormalized BV quantum master equation

$$QI_X[L] + \tfrac{1}{2}\{I_X[L], I_X[L]\}_{g,L} + \hbar\Delta_{g,L} I_X[L] = 0.$$

The BV bracket $\{-,-\}_{g,L}$ and BV Laplacian $\Delta_{g,L}$ are defined by the scale-L heat kernel. We abbreviate this equation to QME. Theorem 3.2 then shows that, for $1 \leq N \leq 3$, amplitudes with primary external fields are computed by the classical cubic interaction with the same propagator. Applying the general framework [4], we obtain the higher genus BCOV invariants on X from the partition function, i.e. from the infrared limit $I_X[\infty]$.

The paper is organized as follows.

Section 1 constructs the quantized local interaction on $\mathbb{C}^N$ from Green functions on marked curves. Stokes' theorem on their moduli space gives the relation among vertices needed for the

QME. Comparing its boundary terms with the short-time limits of heat-kernel integrals, and applying the convergence results of Wang [16], proves the flat-space quantization.

The exact formula of the interaction can be described as follows. In local Betti coordinates for conserved sources $P = (\mathbf{p}_i)$, the theta form on the Jacobian is (see Lemma 1.2)

$$\Theta_g(P) = \sum_{a=1}^{N} \sum_{\mu=1}^{g} \mathrm{d}x_\mu^a \wedge \mathrm{d}y_\mu^a.$$

It determines the coefficients

$$V_{g;\boldsymbol{k}}(P) = \int_{\mathcal{M}_{g,n}} \frac{\Theta_g(P)^{Ng}}{(Ng)!} \prod_{i=1}^{n} \psi_i^{k_i}, \qquad |\boldsymbol{k}| = (3-N)g + n - 3.$$

Replacing each source $\mathbf{p}_i$ by a holomorphic derivative on the ith input gives the local BCOV interaction

$$I(A) = \sum_{\substack{g,n,\boldsymbol{k}\geq 0 \\ 2g-2+n>0}} \frac{\hbar^g}{n!} \operatorname{Tr} V_{g;\boldsymbol{k}}(\partial_{z_1}, \ldots, \partial_{z_n}) \prod_{i=1}^{n} A_{k_i}(z_i)\Big|_{z_1=\cdots=z_n=z} = I_0 + I_+.$$

Here A_{k_i} are the descendant components, I_0 is the genus-zero classical descendant BCOV interaction of Costello–Li [4], and I_+ consists of the positive-genus terms. Tr is the trace map on polyvector fields, given by integration again the trivial holomorphic volume form. Thus the theta-form formula extends the classical interaction to all genera. The construction and normalization are detailed in Section 1.

Section 2 extends the exact formula of the interaction to a compact Calabi–Yau manifold X of arbitrary complex dimension. Section 3 proves the assertion about cubic vertices.

A detailed account of the construction and proofs of the results announced here will appear in a subsequent paper [15]. The above formula suggests an interpretation in terms of the B-twisted topological σ-model coupled to worldsheet gravity. We leave this for future investigation.

All series are formal in genus, arity, and descendant degree. Convergence refers to each coefficient integral. The total genus of a connected graph is $G = b_1(\Gamma) + \sum_v g_v$, where g_v is the genus of the vertex at v. All products and commutators use the total grading.

Acknowledgements. The authors would like to thank Zhengping Gui, Xun Lin, Nicolai Reshetikhin, Junrong Yan, and Shing-Tung Yau for helpful discussions and valuable suggestions. S. L. is supported by NSFC No.12571068. The authors are grateful to ChatGPT 6 Astra for valuable assistance with the exploration of some of the proof strategies and calculations. The authors take full responsibility for the mathematical content and exposition of this paper.

1 Quantization on flat space

In this section, we construct the quantum interaction of BCOV theory on $\mathbb{C}^N$ using Green's functions on marked Riemann surfaces. The interaction coefficients are expressed as integrals over the moduli spaces of these surfaces. We then prove that the resulting effective interaction satisfies the quantum master equation. The proof uses two applications of Stokes' theorem. The first, on the moduli spaces of marked Riemann surfaces, gives relations among the interaction vertices through boundary degenerations. We identify these boundary contributions with the short-time boundary terms of heat-kernel integrals. The second application, on Schwinger parameter space, then gives the quantum master equation.

1.1 Fields and propagators

We recall Costello–Li's descendant formulation [4] of BCOV theory, with their descendant parameter t replaced by $-\zeta$ for later convenience. Set

$$\mathrm{PV}^{i,j}(\mathbb{C}^N) = \Omega^{0,j}(\mathbb{C}^N, \wedge^i T^{1,0}), \qquad \mathcal{E} = \mathrm{PV}(\mathbb{C}^N)[[\zeta]][2], \qquad Q_0 = \bar\partial - \zeta D,$$

where $D = \partial_\Omega$ is the divergence operator for $\Omega = \mathrm{d}z^1 \cdots \mathrm{d}z^N$. We assign cohomology degree $|\zeta| = 2$, $|\hbar| = 6 - 2N$. Write

$$A(\zeta) = \sum_{k\geq 0} \zeta^k A_k, \qquad A_k \in \mathrm{PV}(\mathbb{C}^N).$$

A_0 is the primary field and A_k has descendant index k. An element of $\zeta^k \mathrm{PV}^{i,j}[2]$ has cohomology degree $i + j + 2k - 2$.

Let θ_a be the polyvector generators and $\upsilon_a = \mathrm{d}\bar z^a$. The divergence, gauge-fixing operator, and Laplacian are

$$D = \sum_a \partial_{z^a}\partial_{\theta_a}, \qquad q_0 = -\sum_a \partial_{z^a}\partial_{\upsilon_a}, \qquad H_0 = [\bar\partial, q_0] = -\sum_a \partial_{z^a}\partial_{\bar z^a}. \tag{1.1}$$

Odd derivatives are left derivatives.

For compactly supported inputs $\mu \in \mathrm{PV}_c(\mathbb{C}^N)$, we fix the trace map

$$\mathrm{Tr}(\mu) = c_N \int_{\mathbb{C}^N} [\theta_1 \cdots \theta_N \upsilon_1 \cdots \upsilon_N]\mu \, \mathrm{d}^{2N} z, \qquad c_N = \frac{(-1)^{N(N-1)/2}}{\pi^N}. \tag{1.2}$$

Here the bracket $[-]$ extracts the coefficient of $\theta_1 \cdots \theta_N \upsilon_1 \cdots \upsilon_N$, or equivalently, performs Berezin integration with the corresponding ordering convention.

Let E_t be the kernel of the heat operator e^{-tH_0} on polyvectors, including the inverse trace tensor. Its scalar factor is $(\pi t)^{-N} e^{-|z-w|^2/t}$. Define

$$K_t = DE_t, \qquad R_t = Dq_0E_t, \qquad P_{\epsilon,L} = \int_\epsilon^L R_t \, \mathrm{d}t, \qquad \Delta_t = \partial_{K_t}. \tag{1.3}$$

Following the set-up in [3], for a kernel J, let ∂_J contract one unordered pair of primary slots. Thus the effective propagator $P_{\epsilon,L}$ supplies internal edges, and K_t defines the regularized BV Laplacian and its derived BV bracket $\{-,-\}_t$. Vertices may still have arbitrary descendant inputs. The heat-kernel time t in the propagator is called a *Schwinger parameter*. In a Feynman graph, each internal edge e carries its own parameter t_e. The product of their ranges is the *Schwinger parameter space*; with cutoffs ϵ and L, each t_e ranges over $[\epsilon, L]$.

Lemma 1.1 (Heat-kernel homotopy [3])**.** *With heat-time forms placed before field variables,*

$$[Q_0, \partial_{P_{\epsilon,L}}] = \Delta_\epsilon - \Delta_L, \qquad ([Q_0, -] + \mathrm{d}_t)(\Delta_t - \mathrm{d}t\, \partial_{R_t}) = 0. \tag{1.4}$$

For a local interaction J, the connected renormalization group flow is defined by [3, 4]

$$e^{W(P,J)/\hbar} = e^{\hbar\partial_P} e^{J/\hbar}. \tag{1.5}$$

Its connected graphs carry the usual factor $1/|\operatorname{Aut}(\Gamma)|$ and contain at least one original interaction vertex. A vertex of genus g_v contributes $\hbar^{g_v}$, so the total power is $\hbar^{b_1(\Gamma)+\sum_v g_v}$. Graphs with no external legs are included.

1.2 Green forms and the local interaction

Let $(C; z_1, \dots, z_n) \in \mathcal{M}_{g,n}$, with $2g-2+n>0$. Choose smooth Hermitian metrics h_i on the cotangent lines L_i over $\overline{\mathcal{M}}_{g,n}$ [11], using the conventions

$$\mathrm{d}^c = \frac{i}{2\pi}(\bar\partial - \partial), \qquad \psi_i = -\mathrm{d}\mathrm{d}^c \log \|\ell_i\|_{h_i}.$$

Here ℓ_i is a local nonzero holomorphic frame. The operators d, d^c differentiate the curve and its marked points. On the other hand, we use d_C for the de Rham differential on a single curve C. These differentials do not apply to the sources introduced below.

For a conserved source, we mean a set of vectors $P = (\mathbf{p}_i)$ in $\mathbb{R}^N$ satisfying $\sum_i \mathbf{p}_i = 0$. Let u_P be the vector-valued Green potential on the curve C, unique modulo constants, satisfying

$$\mathrm{d}_C * \mathrm{d}_C u_P = 2\pi \sum_i \mathbf{p}_i \delta_{z_i}, \qquad \sum_i \mathbf{p}_i = 0.$$

In a coordinate w_i centered at z_i, write $u_P = \mathbf{p}_i \log|w_i| + u_i^{\mathrm{reg}} + o(1)$ and put $a_i = \|\mathrm{d}w_i|_{z_i}\|_{h_i}$. Define

$$F_h(P) = \frac{1}{2}\sum_i \mathbf{p}_i \cdot u_i^{\mathrm{reg}} + \frac{1}{2}\sum_i |\mathbf{p}_i|^2 \log a_i, \tag{1.6}$$

and

$$L_P = \frac{1}{2}\sum_i |\mathbf{p}_i|^2 \psi_i, \qquad \beta = \mathrm{d}^c F_h, \qquad \Theta_g(P) = L_P + \mathrm{d}\beta. \tag{1.7}$$

The Euclidean pairing is extended bilinearly when the sources are complexified.

The theta form $\Theta_g(P)$ serves as a basic building block in our construction of the quantum interaction. We give another useful characterization of this form in terms of the periods of the Green current. Let

$$j_P = \frac{1}{2\pi} * \mathrm{d}_C u_P$$

be the current associated with the Green potential. Choose locally a symplectic homology basis $(A_\mu, B_\mu)_{\mu=1}^g$, represented by cycles avoiding the marked points and transported continuously as the curve varies. Define

$$x_\mu^a = \int_{B_\mu} j_P^a, \qquad y_\mu^a = -\int_{A_\mu} j_P^a, \qquad 1 \le a \le N, \quad 1 \le \mu \le g,$$

where j_P^a denotes the ath component of j_P.

Lemma 1.2 (Green forms)**.** *The function F_h is well-defined, independent of the coordinates w_i and the additive constant of u_P, and quadratic in P. The form Θ_g is independent of the metrics h_i. In terms of periods (x_μ^a, y_μ^a),*

$$\Theta_g(P) = \sum_{a=1}^N \sum_{\mu=1}^g \mathrm{d}x_\mu^a \wedge \mathrm{d}y_\mu^a, \qquad \mathrm{d}\Theta_g = 0, \qquad \Theta_g^{Ng+1} = 0. \tag{1.8}$$

More generally, if the sources span a space of dimension ρ, then $\Theta_g^{\rho g+1} = 0$ (rank identity).

Remark (Abel–Jacobi interpretation). When $\mathbf{p}_i \in \mathbb{Z}^N$, each divisor $\sum_i \mathbf{p}_i^a z_i$ has degree zero, and (x^a, y^a) modulo integers gives the Betti coordinates of its usual Abel–Jacobi image in $\mathrm{Jac}(C)$. Under the resulting map to the family of products $\mathrm{Jac}(C)^N$, the form $\Theta_g(P)$ is the pullback of the canonical closed two-form whose restriction to each fiber represents the product principal polarization. For general real sources, we use the period formulas above directly.

The Green and period conventions agree with [8]. The closedness and rank identity in Lemma 1.2 will be used in the boundary argument later.

For $\boldsymbol{k} = (k_1, \dots, k_n)$ with $k_i \geq 0$, define

$$V_{g;\boldsymbol{k}}(P) = \int_{\mathcal{M}_{g,n}} \frac{\Theta_g(P)^{Ng}}{(Ng)!} \prod_i \psi_i^{k_i}, \qquad |\boldsymbol{k}| = (3-N)g + n - 3. \tag{1.9}$$

The integral is an orbifold integral. Unstable data and incorrect degrees give zero. The degree condition $|\boldsymbol{k}| = (3-N)g + n - 3$ follows from $\dim_{\mathbb{C}} \mathcal{M}_{g,n} = 3g - 3 + n$.

Replace each $\mathbf{p}_i$ by a holomorphic derivative on the ith input to set

$$I(A) = \sum_{\substack{g,n,\boldsymbol{k}\geq 0 \\ 2g-2+n>0}} \frac{\hbar^g}{n!} \operatorname{Tr} V_{g;\boldsymbol{k}}(\partial_{z_1}, \dots, \partial_{z_n}) \prod_i A_{k_i}(z_i)\Big|_{z_i=z} = I_0 + I_+. \tag{1.10}$$

Here I_+ includes the powers of $\hbar$ in positive genus. Since the coefficients are defined on $\sum_i \mathbf{p}_i = 0$, a polynomial representative is chosen before substitution. Different representatives differ by a total holomorphic derivative and give the same integrated local functional. The convergence and polynomiality needed for this definition are stated in Lemma 1.6 below.

Lemma 1.3 (Classical coefficients)**.** *At genus zero,*

$$V_{0;\boldsymbol{k}} = \frac{(n-3)!}{\prod_i k_i!}, \qquad I_0 = \sum_{n\geq 3} \frac{1}{n!} \sum_{|\boldsymbol{k}|=n-3} \frac{(n-3)!}{\prod_i k_i!} \operatorname{Tr}(A_{k_1} \cdots A_{k_n}). \tag{1.11}$$

Thus I_0 is the classical descendant BCOV interaction, and its restriction to primary fields is $\operatorname{Tr}(A_0^3)/6$.

The coefficient formula is the genus-zero intersection formula $\int_{\mathcal{M}_{0,n}} \prod_{i=1}^n \psi_i^{k_i} = \binom{n-3}{k_1,\cdots,k_3}$.

1.3 Extension and boundary values of the Green forms

We record the two facts about degeneration needed below: smooth extension of the forms and their restrictions to the boundary.

Let $\widehat{\mathcal{M}}_{g,n}$ be the logarithmic real resolution of $\overline{\mathcal{M}}_{g,n}$ obtained as follows. In a plumbing coordinate, write

$$q_e = e^{-T_e - 2\pi i \tau_e}, \qquad \rho_e = T_e^{-1}, \qquad \tau_e \in \mathbb{R}/\mathbb{Z}.$$

Retain τ_e at $\rho_e = 0$ and resolve the coordinate corners by real blow-ups, beginning with the deepest corners. Thus the boundary records the relative rates of the neck lengths as well as the node angles. This is the conductance version of the resolution in [10, Definition 1.10 and Lemma 1.13]. The construction is understood in orbifold charts.

For a stable graph H with at least one edge, let $\partial_H \subset \widehat{\mathcal{M}}_{g,n}$ denote the corresponding boundary face. Write $E = |E(H)|$, $h = b_1(H)$, and

$$X_H = \prod_v \mathcal{M}_{g_v,n_v}, \qquad g = h + \sum_v g_v, \qquad \lambda_e = \frac{T_e}{\sum_f T_f}.$$

The face is a node-angle torus bundle over $X_H \times \Delta_H^{\mathrm{res}}$, modulo graph automorphisms. Here Δ_H^{res} is the projective length simplex with its relative rates resolved, and its interior is $\lambda_e > 0$, $\sum_e \lambda_e = 1$. Corners describe successive degenerations.

Orient the edges and let P_v be the sum of the original sources at v. The graph current $j^0 = (j_e^0)$ is the unique minimizer

$$j^0 = \arg\min_j \frac{1}{2}\sum_e \lambda_e |j_e|^2, \qquad \sum_{e \text{ outgoing at } v} j_e - \sum_{e \text{ incoming at } v} j_e = P_v. \tag{1.12}$$

At the outgoing and incoming ends of an edge, assign node sources $-j_e^0$ and $+j_e^0$, respectively. Together with the original sources they form a conserved collection P_v' on each component. The two half-edges of a self-loop are treated separately. Set $f_e = |j_e^0|^2/2$.

Lemma 1.4 (Smooth extension)**.** *The forms β, Θ_g, and all their source derivatives extend smoothly to $\widehat{\mathcal{M}}_{g,n}$. The currents j^0 and the functions f_e are smooth on Δ_H^{res}. These extensions are compatible on overlapping plumbing charts and along every nested boundary stratum.*

Proposition 1.5 (Boundary values)**.** *In a local trivialization of the node-angle torus, normalize the cotangent metrics at the node markings by $\|\mathrm{d}w_\alpha\| = 1$. The boundary restrictions are*

$$\begin{aligned} \Theta_g|_{\partial_H} &= \sum_v \Theta_{g_v}(P_v'(j^0)) + \mathrm{d}_\lambda b - \sum_e \mathrm{d}f_e \wedge \mathrm{d}\tau_e, \\ \beta|_{\partial_H} &= b - \sum_e f_e\, \mathrm{d}\tau_e, \end{aligned} \tag{1.13}$$

where

$$b = \sum_v \beta_{g_v}(P_v'(j^0)). \tag{1.14}$$

Here d_λ differentiates the relative lengths. In particular, at fixed λ and node angles,

$$\Theta_g|_{TX_H} = \sum_v \Theta_{g_v}(P_v'(j^0)). \tag{1.15}$$

These formulas may be differentiated in conserved source directions and applied successively at corners.

Lemma 1.6. *The integrals (1.9) converge absolutely and are homogeneous polynomials of degree $2Ng$ in the conserved sources. They are independent of the chosen cotangent-line metrics.*

1.4 Stokes' theorem on curve moduli

We next show that the rank identity in Lemma 1.2 gives a differential identity. This allows us to apply Stokes' theorem and use Proposition 1.5 to evaluate the boundary integral.

For $\sum_i \eta_i = 0$, define

$$D_\eta = \sum_{i,a} \eta_i^a \partial_{p_i^a}, \qquad \Lambda_\eta = \sum_i (\mathbf{p}_i \cdot \eta_i)\psi_i.$$

We first use commuting η_i. All identities are linear in η, so the odd source symbols of the field theory can be substituted afterwards.

Lemma 1.7 (Transgression of the rank identity)**.** *For $|\boldsymbol{k}| = (3-N)g + n - 4$,*

$$\mathrm{d}\left((D_\eta\beta)\frac{\Theta_g^{Ng}}{(Ng)!}\prod_i \psi_i^{k_i}\right) = -\Lambda_\eta \frac{\Theta_g^{Ng}}{(Ng)!}\prod_i \psi_i^{k_i}. \tag{1.16}$$

Proof. Differentiate $\Theta_g^{Ng+1} = 0$ in the direction η and use $D_\eta\Theta_g = \Lambda_\eta + \mathrm{d}(D_\eta\beta)$. The forms Θ_g and ψ_i are closed, and the lemma follows. □

Order the edges $0,\dots,E-1$ and orient the projective simplex by $\mathrm{d}\lambda_1\cdots\mathrm{d}\lambda_{E-1}$. Define the boundary flux

$$\Phi_H=(-1)^{E-1}\sum_{a=0}^{E-1}(-1)^a(D_\eta f_a)\mathrm{d}f_0\wedge\cdots\wedge\widehat{\mathrm{d}f_a}\wedge\cdots\wedge\mathrm{d}f_{E-1}, \tag{1.17}$$

where $\mathrm{d}f_e$ in this formula differentiates only the projective lengths. The current in (1.12) and the flux (1.17) are the only graph data needed for the comparison with heat kernels.

Proposition 1.8 (Boundary identity)**.** *For $|\boldsymbol{k}|=(3-N)g+n-4$,*

$$\sum_i(\mathbf{p}_i\cdot\eta_i)V_{g;\boldsymbol{k}+e_i}(P)=\sum_{\substack{[H,(g_v)]\\E(H)=Nb_1(H)+1}}\frac{1}{|\operatorname{Aut}(H,(g_v))|}\int_{\Delta_H^{\mathrm{res}}}\Phi_H\prod_v V_{g_v;(\boldsymbol{k}_v,0_{\mathrm{int}})}(P'_v). \tag{1.18}$$

The graphs are connected and stable, $g=b_1(H)+\sum_v g_v$, and the external legs are labeled. Each node marking has descendant index zero.

Proof sketch. By Lemma 1.4, Stokes' theorem applies to (1.16) and gives

$$-\int_{\widehat{\mathcal{M}}_{g,n}}\Lambda_\eta\frac{\Theta_g^{Ng}}{(Ng)!}\prod_i\psi_i^{k_i}=\int_{\partial\widehat{\mathcal{M}}_{g,n}}(D_\eta\beta)\frac{\Theta_g^{Ng}}{(Ng)!}\prod_i\psi_i^{k_i}.$$

Use Proposition 1.5 to evaluate the right-hand side. Integration over the node angles selects $E=Nh+1$ and removes $D_\eta b$. With the boundary orientation, the surviving flux is (1.17). The component integrals give the product of vertex coefficients, and the orbifold gluing degree gives the automorphism factor in (1.18). □

1.5 Comparison with the heat-kernel integrals

The short-time collapse of a heat-kernel graph is governed by the same minimizing current j^0 and flux Φ_H as the degeneration of a marked curve. The comparison below identifies its local boundary contribution with the corresponding term in (1.18).

Fix a connected graph H with ordered edges and polynomial holomorphic vertex symbols F_v. Following [16], we place the extended kernel

$$\widehat{K}_{t_e}=K_{t_e}-\mathrm{d}t_e\,R_{t_e}$$

on every edge and apply the vertex operators. Contract using the trace pairing and integrate all vertex positions except one output point. Denote the heat-time degree $E-1$ part by $\omega_H(t;F_v)$: each term has one K edge and $E-1$ propagator terms, with signs determined by the ordered product.

For a fixed upper cutoff L, we use the compactified Schwinger parameter space (see [16, Section 3.1] and [18]). Its added boundary faces record the relative rates at which edge parameters tend to zero. Near the boundary face where all edge parameters vanish, write $t_e=\rho\lambda_e$, with $\sum_e\lambda_e=1$, and let ι_H denote the face $\rho=0$. The local short-time boundary operator is

$$\mathcal{O}_H(F_v)=\int_{\Delta_H^{\mathrm{res}}}\iota_H^*\omega_H(t;F_v). \tag{1.19}$$

This is a local operator before taking the output trace or dividing by the graph automorphism factor. The existence and locality of the boundary restriction follows from [16, Theorems 5–6 and Propositions 9, 11].

The reduced symbol replaces derivatives on the inputs by their even and odd momenta, subject to total momentum conservation. The odd source direction is η, as in (1.17).

Proposition 1.9 (Heat-kernel and curve-boundary comparison)**.** *The operator* (1.19) *vanishes unless* $E = Nh + 1$*. At this edge number its reduced symbol is*

$$\mathrm{Symb}^{\mathrm{red}}\,\mathcal{O}_H(F_v) = \int_{\Delta_H^{\mathrm{res}}} \Phi_H \prod_v F_v(P_v'). \tag{1.20}$$

Here P_v' *is determined by the same minimizing current* (1.12) *as in the curve degeneration. The integral converges on the resolved simplex. The identity also holds for subgraph insertions and successive collapses.*

1.6 Stokes' theorem and the QME

The first Stokes argument and the heat-kernel comparison give local cancellation. The second Stokes argument, on the heat-time parameters, turns this cancellation into the QME at every positive scale.

Proposition 1.10 (Local cancellation)**.** *For the interaction* (1.10)*,* $Q_0 I$ *is cancelled by the sum of the local short-time boundary contributions* $\mathcal{O}_H$*, with their genus powers and graph automorphism factors.*

Proof sketch. The integrated $\bar\partial$ part of a holomorphic vertex is zero, while $-\zeta D$ has reduced symbol

$$\mathrm{Symb}^{\mathrm{red}}(Q_0 I)_{g,\boldsymbol{k}} = -\sum_i (\mathbf{p}_i \cdot \eta_i) V_{g;\boldsymbol{k}+e_i}.$$

Proposition 1.8 expresses its negative as a sum of curve-boundary terms, and Proposition 1.9 identifies each term with the corresponding heat-kernel boundary operator. Integration by parts turns equality of reduced symbols into equality of integrated local functionals. The insertion statement follows from the same comparison; at zero arity, the source flux vanishes. □

Theorem 1.11 (Flat quantization)**.** *The coefficients* (1.9) *define a local interaction on the full descendant field complex. Its effective interaction exists coefficientwise and satisfies*

$$I[L] = W(P_{0,L}, I), \qquad Q_0 I[L] + \tfrac{1}{2}\{I[L], I[L]\}_L + \hbar \Delta_L I[L] = 0, \quad L > 0. \tag{1.21}$$

The statement includes the vacuum coefficient.

Proof sketch. The vertices are well-defined by Lemma 1.6, and the ultraviolet cutoff can be removed coefficientwise by [16]. The homotopy identity in Lemma 1.1 converts the action of Q_0 on the internal edges into a heat-time differential. Apply Stokes' theorem to the heat-time parameters of each connected graph. At the short-time boundary, collapsing subgraphs cancel the action of Q_0 on the vertices by Proposition 1.10. At scale L, cutting an edge gives the BV bracket if the graph disconnects and the BV Laplacian otherwise. These boundary terms give precisely (1.21). □

2 Quantization on a Calabi–Yau manifold

Let X be a closed connected Calabi–Yau manifold of complex dimension N, with holomorphic volume form Ω and a fixed Kähler metric g satisfying

$$\frac{\omega^N}{N!} = c\,\Omega \wedge \overline{\Omega}, \qquad c \text{ constant}. \tag{2.1}$$

Condition (2.1) makes g Ricci-flat, hence real analytic in holomorphic coordinates [7, Theorem 6.1]. Normalize so that $\det(g_{i\bar{j}}) = 1$ in holomorphic volume coordinates. The Chern connection ∇ is torsion-free and preserves Ω. The field complex and differential are

$$\mathcal{E}_X = \mathrm{PV}(X)[[\zeta]][2], \qquad \mathrm{PV}^{i,j}(X) = \Omega^{0,j}(X, \wedge^i T^{1,0}X), \qquad Q = \bar{\partial} - \zeta D, \quad D = \partial_\Omega. \tag{2.2}$$

The trace is the global version of (1.2) in volume charts. See [4] for further details.

To define the interaction, let $\mathfrak{g}$ be the Lie algebra of formal volume-preserving vector fields on $\widehat{\mathbb{C}}^N$ with its standard volume form. We use the Chevalley–Eilenberg (CE) complex of $\mathfrak{g}$; see [9, Chapter 1, §3]. The universal CE background B is the tautological $\mathfrak{g}$-valued one-cochain, defined by

$$B(v) = v, \qquad v \in \mathfrak{g}.$$

We regard its values as polyvector fields of degree one, with the odd CE coefficients placed before the odd polyvector generators. Thus B is even in the total grading and satisfies $DB = 0$. Similarly, we write A_k for the tautological field coordinate of descendant degree k: for a field $\alpha = \sum_{k\geq 0} \zeta^k \alpha_k$, we have

$$A_k(\alpha) = \alpha_k.$$

Thus $A(\zeta) = \sum_{k\geq 0} \zeta^k A_k$ denotes the universal field. From the flat interaction $I = I_0 + I_+$, define

$$\mathcal{R}_B A_k = \delta_{k0} B - B A_{k+1}, \qquad \mathcal{R}_B B = 0, \qquad e^{\mathcal{R}_B} A(\zeta) = B + [e^{-B/\zeta} A(\zeta)]_{\geq 0}. \tag{2.3}$$

Here, products such as BA_{k+1} are taken using the wedge product of polyvector fields, $[\,\cdot\,]_{\geq 0}$ retains the nonnegative powers of ζ. Prolonging $\mathcal{R}_B$ to holomorphic jets gives the equivariant interaction

$$\widehat{I}_B = I_0 + e^{\mathcal{R}_B} I_+. \tag{2.4}$$

Its positive-genus part has no term without ordinary inputs. Choose its representative Nor by eliminating an ordinary output momentum and averaging over the output choices. For n ordinary inputs of descendant degrees $\boldsymbol{\ell}$, use weights $\ell_r/|\boldsymbol{\ell}|$ when $|\boldsymbol{\ell}| > 0$, and $1/n$ otherwise. In this representative, every background input carries at least two holomorphic derivatives.

With the one-form coefficient placed first, set

$$\mathcal{R}_{ij} = -(\bar{\partial}\Gamma^k_{ij})\theta_k. \tag{2.5}$$

Replace field derivatives by covariant derivatives and background jets by curvature jets according to

$$\begin{aligned} \partial_{i_1} \cdots \partial_{i_r} A_k &\longmapsto \nabla_{i_1} \cdots \nabla_{i_r} A_k, \\ \partial_{i_1} \cdots \partial_{i_r} B &\longmapsto \nabla_{i_1} \cdots \nabla_{i_{r-2}} \mathcal{R}_{i_{r-1} i_r}, \qquad r \geq 2. \end{aligned} \tag{2.6}$$

Write $\mathrm{Ev}_{\nabla,\mathcal{R}}$ for this replacement. Covariant derivatives act on all tensor indices; contractions use the parallel volume tensors, and graded multiplication precedes the trace. The resulting interaction on X is

$$\boxed{\mathcal{J}_X = I_0 + \mathrm{Ev}_{\nabla,\mathcal{R}} \,\mathrm{Nor}(e^{\mathcal{R}_B} I_+).} \tag{2.7}$$

Theorem 2.1 (Quantization on a Calabi–Yau manifold)**.** *The interaction (2.7) defines a perturbative quantization of BCOV theory on X: at every scale $L > 0$, its effective interaction $I_X[L]$ satisfies the quantum master equation on the full descendant field complex, at all loop orders and including the vacuum coefficient. The ultraviolet cutoff in its heat-kernel Feynman expansion can be removed coefficientwise by the convergence results of Wang–Yan [17, 18].*

Proposition 2.2 (Metric independence)**.** *Fix the complex structure of X and the volume form Ω. For $a \in H^\bullet(\mathrm{PV}(X), \bar\partial)[[\zeta]][2]$, let a_g denote its g-harmonic representative. The limit*

$$I_X[\infty](a) = \lim_{L\to\infty} I_X[L](a_g) \tag{2.8}$$

exists coefficientwise and is independent of the choice of Kähler metric g satisfying (2.1). This includes the vacuum term $I_X[\infty](0)$.

3 The primary sector for dim ≤ 3

Fix (X, Ω, g) as in Section 2. Let $\mathcal{E}_{\mathrm{prim}} = \mathrm{PV}(X)[2]$ be embedded at descendant degree zero, and put

$$I_{\mathrm{cub}}(A_0) = \frac{1}{6}\,\mathrm{Tr}(A_0^3). \tag{3.1}$$

Lemma 3.1 (Primary vertex reduction)**.** *For $N = 1, 2, 3$, write $\mathcal{J}_X = I_0 + \sum_{g\geq 1} \hbar^g \mathcal{J}_{X,g}$. Then*

$$\mathcal{J}_{X,g}|_{\mathcal{E}_{\mathrm{prim}}} = 0 \quad (g \geq 1), \qquad \mathcal{J}_X|_{\mathcal{E}_{\mathrm{prim}}} = I_{\mathrm{cub}}. \tag{3.2}$$

This holds on all primary fields, without imposing $DA_0 = 0$.

Proof. On primary fields, (2.3) reduces to $A_0 \mapsto A_0 + B$. For n ordinary and m background inputs, (1.9) gives

$$n + m = 3 - (3 - N)g = N - (3 - N)(g - 1) \leq N \qquad (g \geq 1),$$

contradicting the rank requirement $n + m \geq N + 1$ from Lemma 1.2. Curvature evaluation preserves these conditions. In genus zero, (1.11) with all descendants zero forces $n = 3$ and gives the coefficient $1/6$. □

Theorem 3.2. *For $N = 1, 2, 3$, only classical cubic vertices contribute to connected Feynman graphs with primary external inputs, at every loop order and including vacuum graphs.*

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